

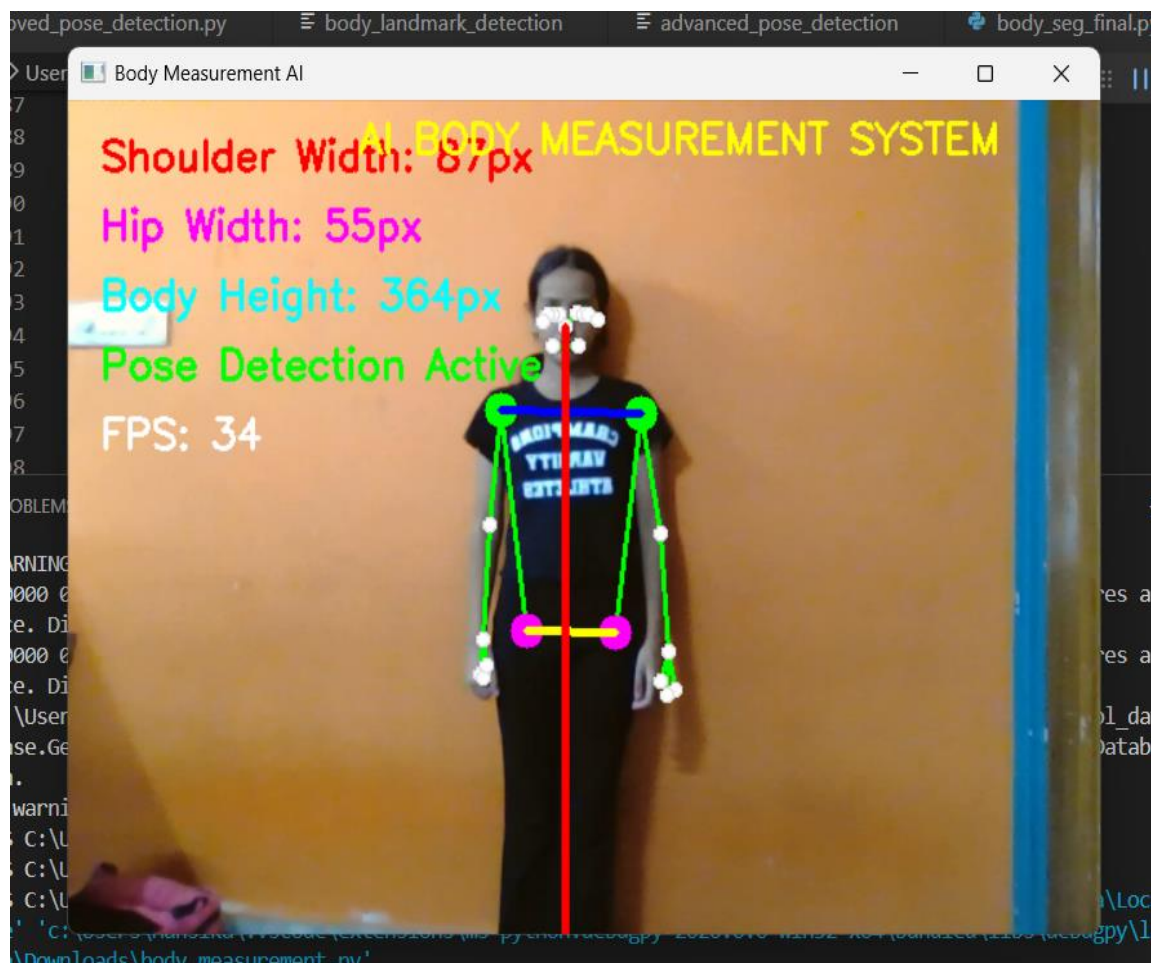
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Role: AI/ML Intern

DAY – 6

Task 1. Improve Body Measurement Accuracy

In this task, I tested the body measurement system using live pose detection and landmark tracking. The AI model successfully detected important body points such as shoulders, hips, and overall body height in real time. Different positions and distances from the camera were tested to improve measurement accuracy. Proper lighting and a straight standing posture produced better results. The system was able to calculate body proportions efficiently, which can help improve virtual try-on accuracy and clothing size estimation



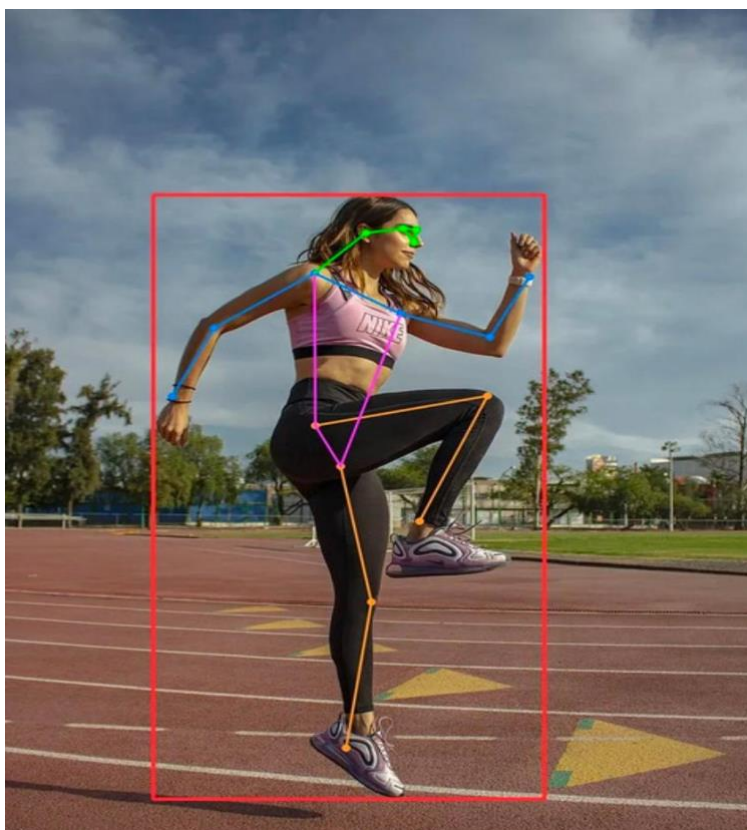
Observation

During testing, the system was able to detect body points like shoulders, hips, and height properly in real time. The measurements were more accurate when the person stood straight in front of the camera with proper lighting. Slight changes in posture and camera distance affected the values a little. The pose detection worked smoothly and gave stable results most of the time. Overall, the testing showed that the system can be useful for virtual try-on and body measurement applications.

Task 2 — Test Pose Detection on Multiple Images/Videos

Pose detection was tested on different images and videos to check the accuracy of body landmark detection. The system successfully detected body points like shoulders, arms, hips, and legs in real time. Better results were observed with proper lighting and clear front-facing poses. Small variations occurred in blurred frames and side angles. Overall, the model worked effectively for pose detection and can be used in body measurement and virtual try-on applications.

Output 1-For Image detection



Output 2-For video detection:



Does Lighting effect Accuracy?

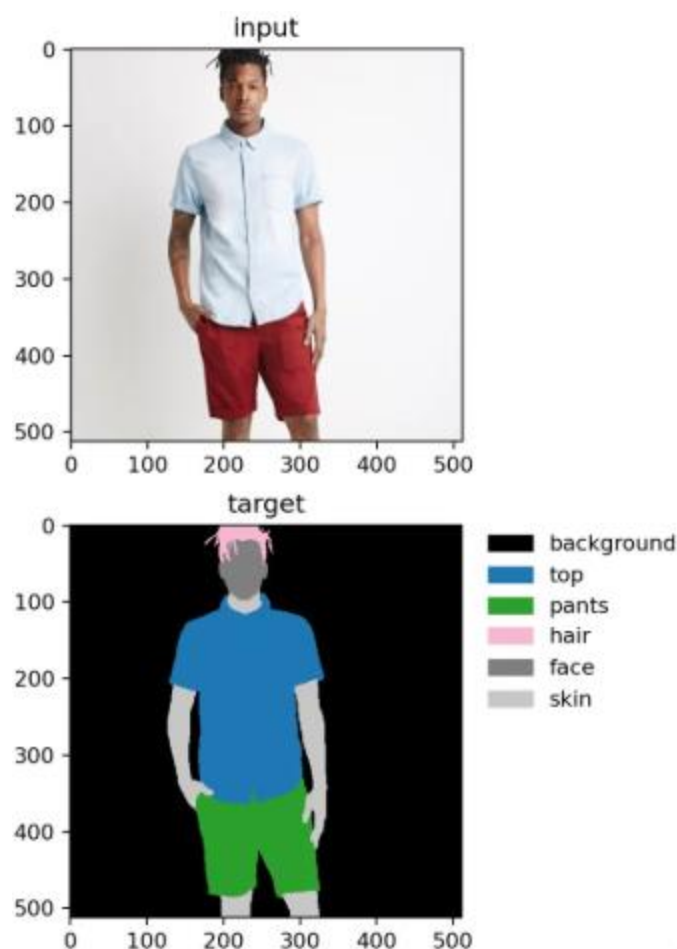
Proper lighting improved pose detection accuracy, while low lighting conditions caused slight variations in body landmark detection.

Is Video Harder Than Audio?

Video-based pose detection was slightly more challenging than static image detection because the model had to track body movements continuously in real time.

Task 3 — Research Cloth Segmentation AI

I researched Cloth Segmentation AI and its use in virtual try-on systems. Cloth segmentation helps identify and separate clothing regions from a person's image using AI and computer vision techniques. This technology is mainly used in online shopping, virtual fitting, and fashion recommendation systems.



Models I researched

- SCHP — most accurate for fashion, detects 20 clothing parts. Best pick for our project.
- SAM (Meta) — you click any region and it instantly masks it. Super flexible.
- U²-Net — fastest, great for background removal before segmentation.
- Mask R-CNN — general purpose, works but not fashion-specific.

What I found useful? -SAM is the easiest to test in Collab

Task 4 — Create Basic Recommendation Dataset

I created a basic recommendation dataset for the AI clothing recommendation system. The dataset included outfit suggestions based on body type, weather, occasion, and style preference. Different clothing combinations were arranged in a simple format so the system can recommend suitable outfits based on user input. During testing, the recommendations changed according to different conditions such as casual wear, formal events, and weather changes.

This task helped in understanding how datasets are prepared and used in AI-based fashion recommendation systems.

Sample Dataset

Body Type	Occasion	Weather	Style	Recommended Outfit
Slim	Casual	Sunny	Modern	T-shirt and Jeans
Athletic	Formal	Cool	Classic	Blazer and Trousers
Average	Party	Winter	Trendy	Hoodie and Denim
Curvy	Office	Rainy	Simple	Kurti and Leggings

Input	Recommended Outfit
Casual + Sunny Weather	T-shirt and Jeans
Formal + Cool Weather	Blazer and Trousers
Party + Winter Weather	Hoodie and Denim
Office + Rainy Weather	Kurti and Leggings
College + Cloudy Weather	Hoodie and Joggers
Wedding + Winter Weather	Kurta and Jacket

In this task I started working on connecting the AI outputs with backend APIs. The body measurement results, pose detection outputs, and outfit recommendations were arranged in a proper format so they can be sent through APIs. I tested how the backend receives user inputs and returns recommendation results. This helped me understand how AI models are connected with real-time applications and how the overall system workflow works.

TASK-5 -- Start integrating AI outputs with backend APIs

In this task, I created a simple backend API demo using FastAPI to understand how AI outputs can be connected with backend services. A basic mock integration was developed where the system accepts user inputs like body shape and occasion, and returns outfit recommendations as API responses. The integration was tested using Swagger UI, which helped in understanding how data is sent and received between the frontend and backend. This task gave practical knowledge about real-time AI application workflow and backend communication.

```
C: > Users > Hansika > Downloads > app.py > recommend
1  from fastapi import FastAPI
2  from pydantic import BaseModel
3
4  app = FastAPI()
5
6  class UserData(BaseModel):
7      body_shape: str
8      occasion: str
9
10 @app.get("/")
11 def home():
12     return {"message": "AI Fashion Recommendation API Running"}
13
14 @app.post("/recommend")
15 def recommend(data: UserData):
16
17     if data.body_shape == "athletic":
18         outfit = "Black T-shirt + Jeans"
19     else:
20         outfit = "Hoodie + Joggers"
21
22     return {"recommended_outfit": outfit}
```

default

GET / Home

POST /recommend Recommend

Parameters

No parameters

Request body required

application/json

Edit Value | Schema

```
{  "body_shape": "athletic",  "occasion": "casual"}
```

Code Details

200

Response body

```
{  "recommended_outfit": "Black T-shirt + Jeans"}
```

Download

Response headers

```
content-length: 46
content-type: application/json
date: Mon, 18 May 2026 12:07:25 GMT
server: uvicorn
```

Responses